3DCANN: A Spatio-Temporal Convolution Attention Neural Network for EEG Emotion Recognition

Shuaiqi Liu, Xu Wang, Ling Zhao, Bing Li, Weiming Hu, Jie Yu, and Yu-Dong Zhang

***Abstract*—Since electroencephalogram (EEG) signals can truly reflect human emotional state, emotion recognition based on EEG has turned into a critical branch in the field of artificial intelligence. Aiming at the disparity of EEG signals in various emotional states, we propose a new deep learning model named three-dimension convolution attention neural network (3DCANN) for EEG emotion recognition in this paper. The 3DCANN model is composed of spatio-temporal feature extraction module and EEG channel attention weight learning module, which can extract the dynamic relation well among multi-channel EEG signals and the internal spatial relation of multi-channel EEG signals during continuous period time. In this model, the spatio-temporal features are fused with the weights of dual attention learning, and the fused features are input into the softmax classifier for emotion classification. In addition, we utilize SJTU Emotion EEG Dataset (SEED) to appraise the feasibility and effectiveness of the proposed algorithm. Finally, experimental results display that the 3DCANN method has superior performance over the state-of-the-art models in EEG emotion recognition.**

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***Index Terms*—3D convolution attention neural network, dual attention learning, EEG emotion recognition, spatio-temporal feature.**

I. Introduction

EMOTION occupies an indispensable position in society and life. Positive emotions can promote and enhance human behavior, while negative emotions can weaken and reduce human behavior. Severe negative emotions, such as depression, will do great harm to human's physical and mental health. Correct recognition of emotions has wide application prospects and practical significance in many aspects [1]–[3] for instance computer science, medical and health, military applications, and affective computing. At present, emotion recognition has turned into an important study direction in emotion computing and medical treatment [4]. In emotion research fields, the recognition methods include two types: verbal acts based on expression and voice and non-verbal acts based on text and physiological signals. Physiological signals are spontaneous and subjective uncontrollability, which can more accurately reflect the real emotional state. EEG [5] is a kind of non-invasive electrophysiological signal, which cannot be disguised. It has a great application prospect in emotion research.

The traditional EEG emotion recognition model is divided into two steps: feature extraction and classification. According to the features of the EEG signal, obtaining the most distinguishing features is vital for feature extraction. The most typical and direct feature extraction approaches are analyzing EEG features separately from the time domain, frequency domain and time-frequency domain. For example, ten time-domain EEG signal features in the task of negative emotion recognition are evaluated in [6] and it is concluded that the first-order difference feature is more helpful to emotion classification. However, the selection of a single feature leads to a poor classification effect of emotional states. In [7], nine time-frequency EEG features jointly input into the random forest classifier for emotion recognition, which significantly optimizes the classification performance of a single feature. When processing the single-channel EEG signal, the combined features of the time domain and frequency domain are input into the neural network to realize emotion recognition [8], thus integrating the advantages of traditional features and neural networks. Many researchers calculate the statistical feature of EEG signals to improve the discriminability of emotional features and achieve efficient emotional recognition. For example, in [9], an emotion classification algorithm founded on the fusion of late positive potential (LPP) and support vector machines (SVM) was proposed, which realized the combination of statistical features and frequency-domain features for EEG processing. In [10], the nonlinear higher-order statistics and neural network technology are used to realize the automatic recognition of EEG emotional signals and achieve good results. With the in-depth development and application of information, signal decomposition, deep learning and other methods have been generally utilized in various fields of EEG signal classification.

EEG is a kind of complex, nonlinear and high time resolution signal. It is hard to express all kinds of details of EEG signal completely only by traditional time-domain feature extraction technology. The signal decomposition can divide the data into different frequencies through different transformations. And the effect of EEG signal classification can be effectively improved by using different processing methods for different frequencies [11]. Popular transforms such as wavelet transform, empirical mode decomposition (EMD) and variational mode decomposition (VMD) have been generally utilized in EEG data processing. Zhang *et al*. [12] put forward an EEG signal feature extraction method ground on EMD and autoregressive model (AR) which verifies the reliability and stability of the emotion recognition. In [13], discrete wavelet transform (DWT) was utilized to obtain information of a small number of EEG channels and frequency bands, and then SVM and k-nearest neighbor classifiers (KNN) were utilized to recognize EEG features which obtained good classification performance. Signal decomposition technology can optimize the performance index of EEG signal classification algorithm based on machine learning. However, the high-level semantic information and multi-level information of EEG signal cannot be obtained by manual feature extraction, while deep learning provides a way to solve this problem.

Traditional machine learning methods require manual extraction of EEG features, which not only have high redundancy in the extraction features but also have poor versatility. Therefore, manual feature extraction methods cannot achieve the desired effect in emotion classification. Obviously, with the continuous development and application of deep learning [14], EEG classification research ground on various neural networks has gradually become the mainstream research direction. Shallow classifiers are difficult to learn deep features, while deep learning has the advantages of strong learning ability and good portability. The biggest difference between deep learning classifiers and shallow classifiers is that deep learning can automatically learn good feature representations instead of manual design. Manually designed features cannot be obtained through big data, even feature extraction time is longer and the accuracy rate is not higher. Zhang *et al*. [15] proposed a spatial-temporal recurrent neural network (STRNN) for emotion recognition and compared the recognition accuracy of SVM and STRNN with differential entropy features. The results show that the performance of STRNN is significantly better than SVM. In [16], the emotion recognition effects of KNN, SVM, decision tree, random forest, and long-short term memory (LSTM) are compared. Among them, LSTM has the best robustness and accuracy.

Therefore, deep learning stands out in the classification of emotional states of EEG signals. For instance, Liu *et al*. [17] studied an original algorithm ground on a multi-layer long short-term memory recurrent neural network (LSTM-RNN) and realized emotion recognition based on EEG by temporal attention (TA) and band attention (BA), which effectively improved the classification effect. Liu *et al*. [18] considered the ability of feature representation and adopt a new type of multi-level features guided capsule network (MLF-CapsNet) to classify emotional states, which greatly improves the calculation speed. Deep learning has a variety of network algorithms. Among them, the convolutional neural network (CNN) has the advantages of high computational efficiency and strong generalization ability compared with traditional artificial neural networks. Therefore, CNN is gradually applied to the field of EEG signal processing. For instance, Gao *et al*. [19] studied the gradient particle swarm optimization (GPSO) model to realize the automatic optimization of the CNN model and obtains higher sentiment classification performance.

Although scholars from various fields of study have entered into the field of emotion research and have proposed a variety of algorithms to improve the effect of emotion recognition and also obtained higher and higher accuracy of emotion classification, they have also encountered unavoidable problems. For instance, the research of selecting representative deep feature and time feature information can greatly optimize the classification results of the emotion experiment, and the research of selecting the minimum number of channels to determine the emotional state of the EEG signal without affecting the accuracy can provide a theoretical basis for portable EEG equipment. Since biological time series analysis is time-varying and continuous, EEG signals contain dynamic information flow, which is of great help to the classification of emotional states. Therefore, it is vital for emotion recognition to choose good temporal or spatial features.

Obviously, the spatio-temporal joint features of EEG have better signal representation performance. Hence, the spatio-temporal network model [20] has become a research hotspot in emotion recognition. The performance of CNN mainly relies on the structure and parameter setting of the network. Therefore, researchers usually adopt the way of adjusting network structure to improve the classification performance. The two-dimension convolution neural network (2DCNN) applies the convolution layer to the two-dimension feature map, which can only obtain the spatial information of the EEG signal. The three-dimension convolution neural network (3DCNN) [21] adds the convolution layer to the three-dimension feature map by introducing time dimension, so as to obtain more comprehensive spatio-temporal information of the EEG data. Compared with 2DCNN, 3DCNN can not only guarantee the synchronization of the time dimension but also mine the detailed information of the time and space domain and realize the effective fusion of time and space features. In EEG signal processing, the traditional 2D convolution can only capture the connection between the EEG channel and the amplitude, so the amplitude fluctuation of each time period is ignored. However, using 3D convolution to perform convolution operations on continuous-time EEG signals can capture time-varying fluctuation information and associated information between EEG channels. Compared with methods based on traditional 2D convolution, 3DCNN associates third-dimensional features, thereby improving recognition accuracy and generalization capabilities.

In addition, scientific studies have described that each part of the cerebral cortex has a unique function and each part of the brain is functionally distributed cooperatively. To simulate the above action of the brain, the attention mechanism (AM) [21]–[23] is proposed. AM can learn and acquire the mapping relationship between different brain regions and emotional tasks, so as to highlight the differences in the spatial and temporal features of different emotions. In [22], channel attention-based emotion recognition networks (CAERN) were proposed, and the feature extraction method based on AM was verified to be conducive to the acquisition of emotion recognition features. Jia *et al.* [23] introduced a spatial-spectral-temporal attention 3D dense network (SST-EmotionNet) and studied the relationship between emotional EEG signals and time changes. Therefore, it is feasible for AM to be applied in the field of EEG emotion recognition. In this paper, we construct 3D convolution attention neural network (3DCANN) to classify the EEG emotion. In 3DCANN, 2D EEG data is converted into 3D and the deep spatio-temporal information can be obtained through the 3DCNN network. Then, the spatio-temporal features are combined with the weights of dual attention learning. Finally, the merged features are input into the softmax classifier for classification.

In addition, Petrantonakis *et al.* [21] proved that the brain areas of the cerebral cortex perform their duties. In fact, the changes in the emotional state only have obvious state changes in certain specific brain areas, such as the prefrontal and temporal lobes, which also shows that different brain regions have different degrees of correlation with emotion recognition tasks and choosing a brain region with high correlation has practical application significance. Therefore, we also study how to select channels in emotional state research to reduce the computational load of the emotion recognition model by adopting a small number of EEG channels. The selective EEG channels also offer some scientific help for the innovation of portable EEG equipment.

The other content planning of this paper is: Section II explains the 3DCANN emotion recognition algorithm proposed in this article; Section III introduces data preprocessing and training process of the 3DCANN structure; Section IV evaluates the classification performance of the algorithm; Section V summarizes the proposed method.

II. Methods

Due to the popularization of EEG processing and the development of computer science, emotion research based on CNN and EEG has been gradually developed. We construct a new structure of CNN named 3DCANN. 3DCANN is composed of 3DCNN and AM. Temporal and spatial analysis is performed using 3DCNN and AM is adopted to study the mapping relationship between EEG emotion features and brain regions. Therefore, 3DCANN can realize efficient classification of various emotions. Fig. 1 describes the structure of the 3DCANN algorithm in this paper, where "⊗" represents the product.

The "Raw EEG signal (5s)" in Fig. 1 describes the structure of a sample. Each sample is 3D data, which consists of five

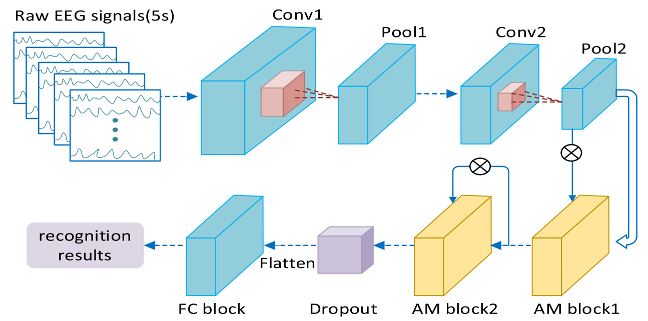

Fig. 1. The flow of the 3DCANN algorithm.

consecutive 1-second 62-channel EEG signals. Fig. 1 describes that 3DCANN has 17 network layers, of which the first four layers are 3DCNN feature extraction structure, the middle layer contains two identical channel attention modules and the last three layers are multilayer perceptron composed of the fully connected layer. The result of the upper layer is mapped to the next layer to realize the valid identification of EEG emotional state. In 3DCANN, we extract spatial features of EEG signal through 3DCNN. We enhance the emotion-related electrodes and suppress the emotion-independent electrodes on the first AM block module by learning the weight matrix. Finally, the high discriminant of high-dimensional EEG features can be achieved by strengthening or weakening the influence of different electrodes of emotion recognition through the second AM block module. Due to the high complexity and small sample size of EEG data, 3DCANN uses a dropout layer with a ratio of 0.2 to prevent the overfitting of the neural network. Finally, the result of the dropout layer imports the Flatten layer and the probability estimation of different emotional states is realized through three fully connected layers. ReLU activation function is adopted in the first two layers of the fully connected layers and the softmax activation function is applied in the last layer to estimate the probability of emotional states. The spatial-temporal feature extraction and attention mechanism of the network are described as following, respectively.

A. Spatio-Temporal Feature Extraction

Deep learning has achieved great success in EEG data processing, including schizophrenia classification [24], Attention deficit/Hyperactivity disorder (ADHD) classification [25] and emotion recognition [26]. It usually contains the convolution layer, pooling layer and other basic structures in a traditional 2DCNN model. The input layer and the output layer are the input and output of the algorithm, respectively. Each neuron in the convolutional layer extracts local features at the same position in several feature maps of the previous layer. Assuming that the convolutional layer is the *i*-th layer, and the *j*-th feature map of this layer is $k_{ij}^{x_0 y_0}$, then the convolutional value at position (x_0, y_0) can be denoted as $k_{ij}^{x_0 y_0}(x_0, y_0)$, which is

$$k_{ij}^{x_0 y_0}(x_0, y_0) = \tanh\left(b_{ij} + \sum_{c}\sum_{p=0}^{P_i-1}\sum_{q=0}^{Q_i-1} w_{ijc}^{pq} k_{(i-1)c}^{(x_0+p)(y_0+q)}\right) \tag{1}$$

where b_{ij} is the bias belonging to the *j*-th feature map. w_{ijc}^{pq} is the weight value at the (p, q) position of the convolution kernel connecting the *c*-th (*c* traverses all the feature maps at the $i-1$ layer connected to the current feature map) feature map at the $i-1$ layer and the *j*-th feature map at the i layer, P_i and Q_i are the number of rows and columns of the convolution kernel, respectively. $\tanh(\cdot)$ is the hyperbolic tangent function. Through the pooling layer, the characteristics of the convolutional layer are subsampled, and the convolutional neural network has a certain scale invariance to reduce the amount of data calculation and retaining useful information. CNN adjusts parameters through supervised or unsupervised methods to obtain the optimal solution of the parameters.

Convolution is applied to two-dimension feature maps which only extract spatial domain features in 2DCNN. The 3D convolution include temporal information is required since it is necessary to capture dynamic information in a continuous-time period when analyzing EEG signals. The 3D convolution can fuse dynamic and detailed information of EEG signals which can learn more representative spatio-temporal features compared to 2D convolution. Traditional 2D convolution is usually applied to images and learning features only in the spatial dimension will result in low accuracy. 3D convolution is mostly used in the video, and the time dimension is introduced into the convolution kernel, which can learn the dynamic information of continuous frames and effectively improve the classification effect. Therefore, we establish a 3DCNN model to extract the spatial and temporal features of EEG signals. The 3DCNN constructed in this paper contains structures such as 3D convolution layer, 3D pooling layer, 3D dropout layer and fully connected layer, which will be explained one by one as following.

1) 3D Convolution Layer: 3DCNN is suitable for spatio-temporal feature learning. The 3D convolution is performed in time and space, which can make full use of temporal and spatial information of data for modeling. In the 3DCNN model, the feature maps in the convolution layer will be connected to multiple adjacent time points of the previous layer which can obtain deeper time information. Assume that $k_{ij}^{x_0y_0z_0}(x_0y_0z_0)$ is the convolution value at the position (x_0, y_0, z_0) of the *j*-th feature map in *i*-th layer. Then extending 2D convolution to 3D convolution can be expressed as

$$k_{ij}^{x_0y_0z_0}(x_0y_0z_0) = \partial(\mathrm{b}_{ij} + \sum_c \sum_{p=0}^{P_i-1} \sum_{q=0}^{Q_i-1} \sum_{r=0}^{R_i-1} w_{ijc}^{pqr} k_{(i-1)c}^{(x_0+p)(y_0+q)(z_0+r)}) \quad (2)$$

where ∂ is the activation function, R_i is the size of the three-dimension kernel along the time dimension, and w_{ijc}^{pqr} is the weight value at the (p, q, r) position of the convolution kernel connecting the *c*-th feature map of the $i-1$ layer and the *j*-th feature map of the *i*-th layer. We use the ReLU in the convolution layer in this paper, and its calculation formula is

$$ReLU\,(k) = \phi\,(k) = \max\,(0, k) \quad (3)$$

2) 3D Pooling Layer: The common pooling layer [27] consists of max pooling and average pooling in neural networks. Pooling can reduce the size of spatial features and the risk of overfitting which plays a role in compressing feature maps. The two-dimension max pooling is to place the window of 2×2 at the non-overlapping position of each feature map and take the maximum value of each window as the output. Let the *l*-th layer be a two-dimension max pooling layer, $s_{l,m} \in Q^{\dim(s_{l,m})_1 \times \dim(s_{l,m})_2}$ be the *m*-th feature map in the *l*-th layer. The *m*-th feature map in the previous layer can be denoted as $s_{l-1,m} \in Q^{2\dim(s_{l,m})_1 \times 2\dim(s_{l,m})_2}$. Therefore, the element of the m-th feature map in the *l*-th layer at position (x_1, x_2) is defined as $s_{l,m}^{x_1,x_2}$, which can be calculated as

$$s_{l,m}^{x_1,x_2} = \max(Q_{l,m}^{x_1,x_2}) \quad (4)$$

$$Q_{l,m}^{x_1,x_2} = \{s_{l-1,m}^{y_1,y_2} | y_1 \in \{2x_1-1, 2x_1\}, y_2 \in \{2x_2-1, 2x_2\}\} \quad (5)$$

where the $Q_{l,m}^{x_1,x_2}$ set contains all the elements in the 2×2 window. The *l*-th layer obtains the maximum value of the set as output through the max function. The two-dimension max-pooling layer down-samples the features map of the $l-1$ layer, and finally realizes the compression of the entire feature map. Similar to the high-dimension convolution expansion method, the two-dimension max-pooling can also be extended to three-dimension max-pooling.

Let $s_{l,m} \in Q^{\dim(s_{l,m})_1 \times \dim(s_{l,m})_2 \times \dim(s_{l,m})_3}$ denote the *m*-th feature map in the *l*-th layer, and the *m*-th feature map of the previous layer is denoted as $s_{l-1,m} \in Q^{2\dim(s_{l,m})_1 \times 2\dim(s_{l,m})_2 \times 2\dim(s_{l,m})_3}$. Therefore, the element $s_{l,m}^{x_1,x_2,x_3}$ at position (x_1, x_2, x_3) of the *m*-th feature map in the *l*-th layer can be obtained as

$$s_{l,m}^{x_1,x_2,x_3} = \max(Q_{l,m}^{x_1,x_2,x_3}) \quad (6)$$

$$Q_{l,m}^{x_1,x_2,x_3} = \{s_{l-1,m}^{y_1,y_2,y_3} | y_i \in \{2x_i-1, 2x_i\}, i \in 1,2,3\} \quad (7)$$

where the $Q_{l,m}^{x_1,x_2}$ collection contains all the elements in the $2 \times 2 \times 2$ window.

3) 3D Dropout Layer: Dropout [28] refers to deactivating neurons with a certain probability, which not only reduces the computational burden of the network but also optimizes the adaptability of the convolution neural network. It is a common method used to prevent overfitting. Let the *l*-th layer be the 3D dropout layer, the *m*-th feature in *l*-th layer is computed as $s_{l,m} \in Q^{\dim(s_{l,m})_1 \times \dim(s_{l,m})_2 \times \dim(s_{l,m})_3}$ and the *m*-th feature in the $l-1$ layer can be denoted as $s_{l-1,m} \in Q^{\dim(s_{l,m})_1 \times \dim(s_{l,m})_2 \times \dim(s_{l,m})_3}$. Then the 3D dropout layer can be defined as

$$s_{l,m} = g_{l,m} s_{l-1,m} \quad (8)$$

$$g_{l,m} \sim B(1, 0.5) \quad (9)$$

where $g_{l,m}$ is the threshold corresponding to the *m*-th feature tensor. The 3D Dropout layer can closely follow other layers to prevent overfitting of the deep network.

4) Fully Connected Layer: Fully connected layers are generally connected after multiple convolution layers and pooling

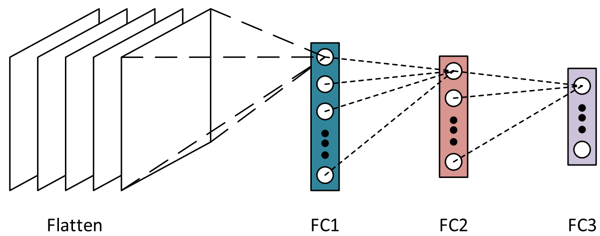

Fig. 2. Structure diagram of multilayer perceptron.

layers in 3DCNN. Let layer l be the fully connected layer, and a multilayer perceptron [29] is formed if the previous layer is also a fully connected layer as described in Fig. 2, where FC represents the fully connected layer.

If the previous layer is not a fully connected layer, then the input of the l-th layer should be the feature map of the output from the $l-1$ layer. In this paper, the activations of the l-th layer and the $l-1$ layer are denoted as $s_l \in Q^n$ and $s_{l-1} \in Q^m$ respectively, so the fully connected layer of the l-th layer can be defined as

$$s_l = \tanh(f(s_{l-1})) \tag{10}$$

$$f(h) = Wh + b \tag{11}$$

where the function f is the full connection operator. The linear transformation $W \in R^{n\times m}$ is used to transform the m-dimensional vector into the n-dimensional vector, while bias is denoted as $b \in R^n$. In 3DCANN, the last fully connected layer determines the emotional state corresponding to EEG signals through softmax activation function, which can be expressed as

$$p_k = \frac{\exp(w_k^T x)}{\sum_j \exp(w_j^T x)} \tag{12}$$

where x represents a sample, k represents an emotional state. $\sum_j w_j^T x$ is the normalization term and the sum of the guaranteed probabilities is 1. The softmax classifier has unique advantages when dealing with natural language, multi-dimension signals and color image data, and is popularized in deep learning models.

B. Channel Weight Learning Based on Dual Attention

Various regions of the brain, for example, the prefrontal and temporal brain regions, occupy different weights in the process of emotion recognition. So, the enhancement or suppression of a certain brain region will affect the effect of EEG emotion recognition. The attention mechanism can learn the weight map of different EEG channels that affect the effect of emotion recognition. AM can enhance the key emotional information in EEG signals and suppress useless information that interferes with the recognition effect. In 3DCANN, the attention mechanism (AM block, as shown in Fig. 1) is added to the constructed neural network to calculate the channel weight coefficient matrix to enhance the EEG channels with high correlation and suppress the EEG channels with low correlation. The attention module of this study is described as Fig. 3, which includes two permute layers, a dense layer and a multiply layer. The dense layer uses softmax to learn the attention weight matrix of the channel, and

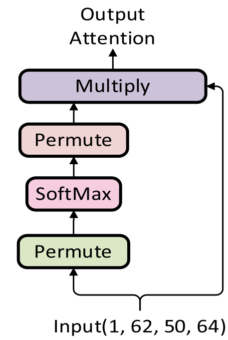

Fig. 3. Structure of AM block.

the multiply layer realizes the fusion of AM block input and weight matrix.

Let h denote the set of AM block input, which can be defined as:

$$h = \{h_1, h_2, \ldots, h_D\} \tag{13}$$

where D refers to the D-th EEG channel. The elements in each set represent the input information corresponding to a certain channel. The permute layer can realize the dimensional conversion of the set h to obtain the set h'. After passing the set h' through the softmax function in the dense layer, the channel attention weight matrix α' is obtained, which can be expressed as

$$\alpha'_i = \frac{\exp(h'_i)}{\sum_{j=1}^{D} \exp(h'_i)} \tag{14}$$

The dimensional conversion of matrix α' is achieved through the permute layer to obtain matrix α. Finally, the weight matrix α and the set h are merged through the product. Therefore, the feature map v which is processed by the attention module can be expressed as

$$v = \alpha h \tag{15}$$

To maximize the advantages of channel attention weights, we adopt the dual attention module structure [30] to obtain more discriminative spatio-temporal features of EEG signals. The dual attention module contains two identical attention modules. The first attention module learns the channel attention weight matrix and merges the matrix with the input to obtain a new feature map, while the second attention module is responsible for enhancing the channel weight results obtained by the first attention module, and finally output more effective features for classification by the classifier.

The 3DCANN model constructed in this paper can well extract the spatio-temporal features, which is helpful to understand EEG signals from the perspective of time and space. We will give a detailed introduction to the training of the model later.

III. Experiments

A. Data and Preprocessing

To evaluate the stability and feasibility of the 3DCANN structure, we adopt the SJTU Emotion EEG dataset public

TABLE I
THE DESCRIPTION OF THE SEED DATASET

Type	Description
Subjects	15 (7 males and 8 females)
Channels	62
Sampling rate	200Hz
Movies	6 different movie clips
Session	3, There are 15*15 experiments per session

TABLE II
PARAMETERS SETTING OF 3DCANN

Layer	Type	Output shape	Kernel size
1	Conv3D	5×62×200×32	5×5×5
2	MaxPooling3D	2×62×100×32	2×1×2
3	Conv3D	2×62×100×64	5×5×5
4	MaxPooling3D	1×62×50×64	2×1×2
5	Permute	1×64×50×62	-
6	Dense	1×64×50×62	-
7	Permute	1×62×50×64	-
8	Multiply	1×62×50×64	-
9	Permute	1×64×50×62	-
10	Dense	1×64×50×62	-
11	Permute	1×62×50×64	-
12	Multiply	1×62×50×64	-
13	Dropout	1×62×50×64	-
14	Flatten	198400	-
15	FC1	1024	-
16	FC2	512	-
17	FC3	3	-

emotion recognition database to test the proposed model. The SEED dataset collects EEG signal data by the way that subjects watch a 4-minute movie to trigger emotions. The collected data includes three emotional states: positive, neutral and negative. Each subject was given a mood test in three different time periods and a total of 45 (15*3) experiments were conducted. Each experiment includes 62 channels of EEG emotion data for 4 minutes. The specific details of the SEED dataset are described in Table I. In Table I, Session represents 3 different periods of the EEG experiment. In each period, each subject performed 15 times of EEG experiment of watching movies.

The EEG signals in all channels have a large amount of noise information that has nothing to do with emotions. If the original signals are used directly, not only the characteristic information of the real EEG signals cannot be extracted, but also interfere with experimental results and affect the ability to recognize categories. In this study, we adopt EEGLAB to preprocess the original EEG signal, for instance, deleting electrooculogram (EOG), electromyography (EMG) and power frequency interference, and then conducting EEG emotion experiments. The processing steps of EEGLAB can be divided into 4 steps. First, the EEG signal of 0.3∼49Hz is retained through the FIR filter. Second, the average value of the surrounding electrodes is used for interpolation to obtain the waveform of the abnormal electrode. Third, remove signals that are heavily polluted by noises such as EMG. Finally, the independent component analysis algorithm is used to remove noises such as blinking. In this study, the EEG signals between [0.5, 60.5] seconds of each experiment are selected for non-overlapping segmentation through the 1-second sliding window, and then 2D data were selected for continuous 5 seconds to be merged into 3D data, and finally 3D data were input into the 3DCANN network for training and testing.

B. Model Training

The 3D convolution attention neural network includes two parts: 3DCNN feature extraction and EEG channel attention weight learning. The spatial and temporal characteristics of different emotional states were obtained by 3DCNN and the channel attention is used to learn the weight matrix of different electrodes affecting the emotional state. The proposed algorithm is implemented on the TensorFlow 2.0 platform. The hardware environment is Windows Server 2008 R2 Standard operating system, 32G RAM, Intel(R) Xeon(R) CPU E5-2667, NVIDIA Tesla K40c GPU and Matrox G200eR GPU. The specific training details of the 3DCANN model are as follows.

The 5 seconds of the 2D EEG is adopted as input, and the size of the input matrix is $5 \times 62 \times 200 \times 1$ (time, number of electrodes, sampling points). The 3DCNN feature extraction part includes two convolution layers. Each convolution layer is connected to a max-pooling layer, the convolution kernel size is $5 \times 5 \times 5$, the activation function is ReLU, the window size of the first pooling layer is $2 \times 1 \times 2$, the window size of the second pooling layer is $2 \times 1 \times 2$ and the max-pooling layer can reduce the feature map size and network parameters. The attention module includes two permute layers, a dense layer and a multiply layer. The activation function of the dense layer is softmax. In order to prevent over-fitting, the network also has a dropout layer and a flatten layer. The specific parameter settings of the model are shown in Table II. In Table II, "-" indicates that the value does not exist, and "FC1, FC2, FC3" indicate different fully connected layers respectively.

In this study, the adaptive moment estimation (Adam) optimizer is used for model optimization. To prevent the small initial stage accumulation caused by zero initialization, a deviation correction operation is also carried out in Adam. We minimize the cross-entropy loss function through the Adam optimization method, which is:

$$L(p, q) = -\sum p(x) \log q(x) \tag{16}$$

where p is the true label and q is the probability of prediction.

It is worth noting 2D EEG is turned to 3D EEG by the sliding window which retains the temporal information of EEG signals to the maximum extent. Combining the spatio-temporal feature extraction characteristics of 3DCNN's and attention mechanism, the 3DCANN method effectively optimizes the comprehensiveness of EEG features and reflects the relationship between different channels and emotional states.

IV. RESULTS AND DISCUSSION

To accurately describe the reliability of the 3DCANN model, we conduct subject-independent EEG emotion recognition experiment and subject-dependent EEG emotion recognition experiment. And we also carry out statistical analysis on the

TABLE III
THE PERFORMANCE OF EACH ALGORITHM IN SUBJECT-INDEPENDENT EXPERIMENTS

Method	Accuracy
DGCNN [31]	79.95%
CNN-DDC [32]	82.10%
Fea-SVM [33]	83.33%
BiDANN [34]	83.28%
R2G-STNN [35]	84.16%
BiDANN-S [36]	84.14%
RGNN [37]	85.30%
3DCANN (Our method)	**96.37%**

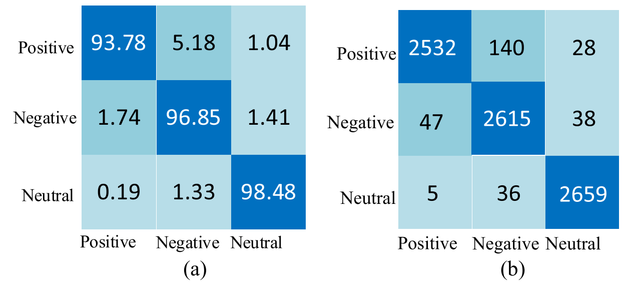

Fig. 4. Confusion matrix of the subject-independent experiment. (a) the percentage of confusion matrix (%), (b) Number of samples for the confusion matrix.

results of the EEG channel selection experiment. For subject-independent EEG emotion recognition, we apply leave-one-subject-cross-validation, that is, each time one subject is used as the test set and the other 14 subjects as the training set. The training data and test data are from different subjects in the leave-one-subject-cross-validation. We use 15-fold cross-validation to achieve subject-dependent EEG emotion recognition, that is, the data of all subjects are divided into 15 parts evenly, and one part is used as the test set and the other 14 parts as the training set each time. In this cross-validation, the training data and test data are from the same subject. Furthermore, we study the EEG channels related to emotional tasks through the distribution of attention weights and compare the recognition results of key channels with full channels.

A. Subject-Independent Emotion Recognition Experiment

To comprehensively study the emotion recognition effect of the 3DCANN, we present several state-of-the-art algorithms. These algorithms are: (1) An emotion recognition method based on dynamic graph convolutional neural networks (abbreviated as DGCNN) proposed in [31]. (2) A cross-subject EEG classification research based on convolutional neural network and deep domain confusion (abbreviated as CNN-DDC) proposed in [32]. (3) An algorithm that combines 18 linear and non-linear EEG features and applies SVM (abbreviated as Fea-SVM) for efficient cross-subject emotion recognition proposed in [33]. (4) EEG emotion recognition method based on bi-hemispheres domain adversarial neural network (abbreviated as BiDANN) proposed in [34]. (5) An emotion recognition method based on the spatio-temporal neural network model with a regional to the global hierarchical feature learning process (abbreviated as R2G-STNN) proposed in [35]. (6) The EEG emotion detection method is based on the improved version of the bi-hemisphere domain adversarial neural network model (abbreviated as BiDANN-S) proposed in [36]. (7) The emotion detection based on regularized graph neural network (abbreviated as RGNN) proposed in [37]. The comparison results of the various methods are presented in Table III.

Table III describes that the recognition accuracy of the 3DCANN structure in this paper is as high as 96.37%, which is currently the best algorithm in all subject-independent emotion recognition algorithms. From Table III, we find that the accuracies of DGCNN, CNN-DDC, R2G-STNN and other methods are not high, which reflects that the two-dimension mode cannot well capture the temporal and spatial discriminative features, while the three-dimension mode can express the dynamic changes of the signal. In addition, the Fea-SVM algorithm in the table does not work well, which verifies that traditional machine learning methods cannot automatically extract feature representations and feature information is not rich enough. 3DCANN can obtain high recognition accuracy since it not only effectively extracts the time change information of emotional state, but also extracts the spatial relationship information between each electrode, which is conducive to fully learning high discriminative spatio-temporal features and solves the problem of high-dimension feature redundancy.

In addition, the percentage of confusion matrix of the three emotional states based on the 3DCANN algorithm is represented in Fig. 4(a) to visually represent the confusion effects of different emotions. Fig. 4(b) shows the sum of test samples for leave-one-subject-cross-validation corresponding to Fig. 4(a). In Fig. 4(b), each row represents the actual number of samples in the category, and each column represents the number of samples predicted to be the category. It can be seen that 3DCANN can effectively classify emotions from Fig. 4. In Fig. 4, the probabilities of our algorithm incorrectly identifying positive, negative and neutral emotion samples is 6.22%, 3.15% and 1.52%, respectively. Because the positive emotion samples of subject 1 were misrecognized more frequently, the average accuracy of positive emotions is slightly lower than that of other emotions. This may be due to the large difference between the EEG signals of subject 1 and other subjects. In particular, the average accuracy of the three emotions is above 93%, achieving a good recognition effect, indicating that 3DCANN has good generalization ability.

To fully demonstrate the recognition ability of the 3DCANN algorithm, the receiver operating characteristic (ROC) curve and area under the ROC curve (AUC) of all subjects are calculated. ROC directly illustrates the advantages and disadvantages of sensitivity and specificity indexes. AUC is the area under the ROC curve, which can directly observe the performance of the classifier. The larger the value of AUC is, the better performance is. In this paper, the three-class ROC curve is obtained by the micro-average measurement method. First, the test sample

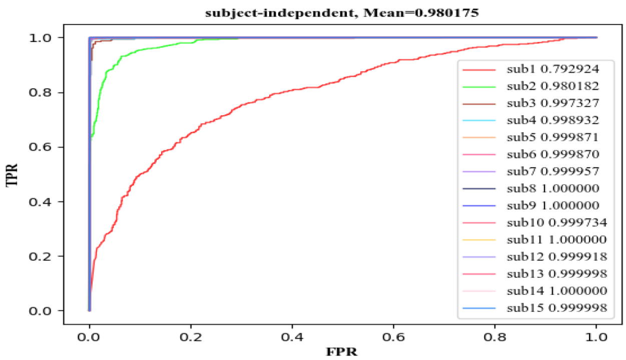

Fig. 5. Micro-average ROC curve of subject-independent experiment.

includes two characteristics: (1) Each row of the label matrix marks a category to which the test sample belongs. The label is only composed of 0 and 1, where 1 represents the correct emotion category and 0 represents other emotion categories. (2) Each row of the probability matrix represents the probability value of a test sample in each category. If a sample is correctly identified, the position of 1 in the sample label corresponds to the position of the largest element in the row of the probability matrix. Then, according to these two characteristics, the label matrix and the probability matrix are expanded in rows and transposed to obtain two columns to form a two-class result. Finally, the final three-class micro-average ROC curve is obtained.

Fig. 5 describes the ROC curve and AUC value of the leave-one-subject-cross-validation results of 3DCANN in the subject-independent experiment, where sub1 represents the ROC curve and AUC value of subject 1 as the test set and the other subs are the same. In particular, although the accuracy rates of subjects 1 and 2 are 62.96% and 89.81%, respectively, their AUC values reached 0.79 and 0.98, and the average AUC value reached 0.98, indicating that the proposed algorithm was robust and conducive to EEG emotion classification. The distribution of leave-one-subject-cross-validation was not even, which may be due to the degrees of difference in the emotional characteristics of EEG signals of subjects that vary in the same scene.

B. Subject-Dependent Emotion Recognition Experiment

Similar to the subject-independent emotion recognition experiment, we introduce several advanced algorithms. The algorithms are: (1) An EEG emotion recognition method based on electrode-frequency distribution maps (abbreviated as EFDMs) proposed in [38]. (2) An emotional state research technique based on the deep belief network (abbreviated as DBN) proposed in [39]. (3) An emotional state research experiment based on functional connectivity patterns (abbreviated as FCP) was proposed in [40]. (4) The EEG emotion classification algorithm based on spiking neural networks (abbreviated as SNNs) was introduced in [41].

In order to better reflect the performance advantages of 3DCANN, this paper lists the above methods from the perspective of deep spatiotemporal networks and traditional emotion recognition algorithms. Compared with deep neural networks such as DGCNN and RGNN, the 3DCANN algorithm has outstanding performance in the field of emotion recognition due to its advantages in three-dimension mode and channel attention. Compared with traditional emotion recognition algorithms such as EFDMs, it verifies that deep spatiotemporal networks are more conducive to the research of emotion recognition. In addition, the above methods are all subject-dependent emotion recognition methods, reflecting the research results of the training set and test set from the same subject, which is consistent with the experimental purpose of this section. The evaluation indexes of each method are described in Table IV.

TABLE IV
THE PERFORMANCE OF EACH ALGORITHM IN SUBJECT-DEPENDENT EXPERIMENTS

Method	Accuracy
DGCNN [31]	90.40%
EFDMs [38]	90.59%
DBN [39]	92.87%
BiDANN [34]	92.38%
RGNN [37]	93.38%
FCP [40]	95.08%
SNNs [41]	96.67%
3DCANN (Our method)	**97.35%**

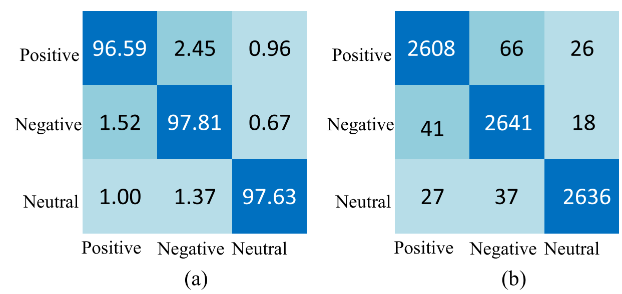

Fig. 6. Confusion matrix of the subject-dependent experiment. (a) the percentage of confusion matrix (%), (b) Number of samples for the confusion matrix.

Table IV illustrates that the recognition accuracy of the 3DCANN algorithm can reach 97.35% and achieve the optimal performance in all of the compared EEG emotion recognition methods. From Table IV, we find that the 3DCANN method is superior to SNNs, FCP and other methods, which may be the reason for introducing the channel attention mechanism to maximize the difference in characteristics of different emotions. However, methods such as EFDMs and DBN are implemented by means of single features and neural networks. To a certain extent, some spatiotemporal information is lost. 3DCANN not only effectively excavates the deep temporal and spatial information of EEG signals, but also effectively weakens the influence of electrodes unrelated to emotion recognition. Fig. 6(a) shows the percentage of confusion matrix obtained by applying 3DCANN in subject-dependent emotion recognition experiment. Fig. 6(b) shows the sum of test samples for 15-fold cross-validation corresponding to Fig. 6(a). In Fig. 6(b), each row represents the actual number of samples in the category, and each column

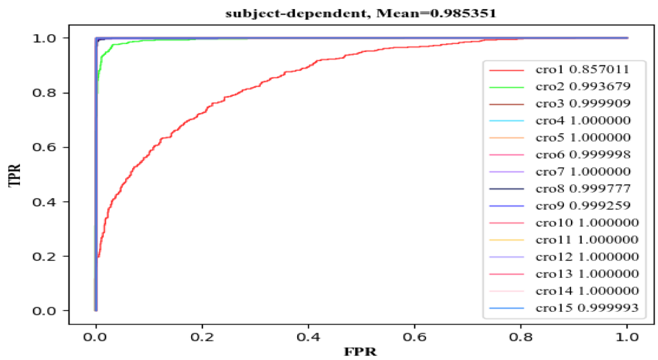

Fig. 7. Micro-average ROC curve of subject-dependent experiment.

represents the number of samples predicted to be the category. It can be observed from Fig. 6 that 3DCANN can effectively classify emotions.

It can be observed from Fig. 6 that the probabilities of incorrect identification of positive, negative and neutral emotion samples of our method are 3.41%, 2.19% and 2.37%, respectively. Since the data of all subjects is randomly scrambled and then cross-validation, the data distribution of different categories in the training data and test data is unbalanced, and the number of samples of different subjects is also quite different. This may be the reason why the correct rate of positive emotion recognition is slightly lower than that of negative and neutral emotions. Although the accuracy of positive emotions is slightly lower, the accuracy of the three emotions is not much different, and all reach more than 96%, indicating that our algorithm has good robustness.

Fig. 7 shows the ROC curve and AUC value of the 15-fold cross-validation result in the subject-dependent emotion recognition experiment, where cro1 represents the ROC curve and AUC value of the first run result in the 15-fold cross-validation, and the other cros are the same. In particular, the accuracy of cro1 is only 67.78% while the AUC value reaches 0.86 and the average AUC reaches 0.985, indicating that the algorithm is robust and able to better identify the human emotional state. Also, the distribution of 15-fold cross-validation was not even, which may be due to the imbalance of the training and test data, that is, the number of samples of different categories varies due to the random shuffled data and cross-validation.

C. Channel Selection Based on Attention Mechanism

We use the attention mechanism to study EEG channels related to emotion in this experiment. To intuitively reflect the influence of channels on emotion recognition, we map the attention weight distribution of AM block1 in Fig. 1 to the brain scalp, as shown in Fig. 8, which describes the topographic maps of the attention weight of subject-independent experiment and subject-dependent experiment. It can be observed from Fig. 8 that the emotional state is highly correlated with brain regions such as the prefrontal lobe, temporal lobe and occipital lobe. This is similar to the results of [42]–[44]. Experiments in [42] show that frontal lobe asymmetry is a regulator of emotion. Experiments

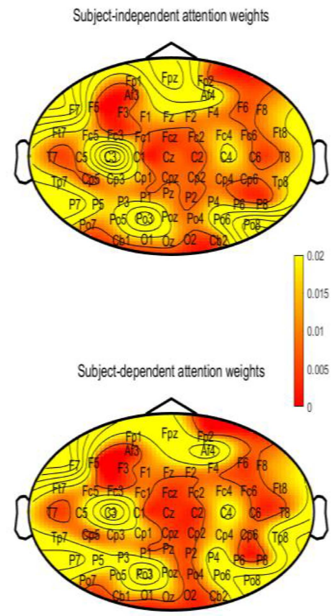

Fig. 8. Attention weight distribution of emotion recognition.

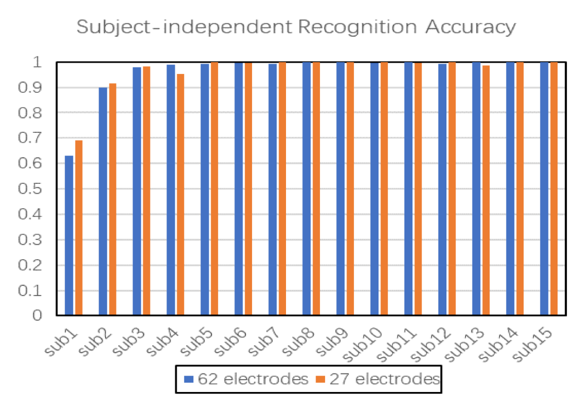

Fig. 9. Recognition results of subject-independent experiment with different EEG channels.

in [44] found that the difference in emotional characteristics is mainly reflected in the occipital lobe and temporal lobe brain regions. Channel selection based on the attention mechanism can not only reduce the amount of calculation but also facilitate the development of portable EEG equipment. Therefore, studying the channel selection of EEG emotion recognition has practical significance.

To study the EEG channels that are useful for emotional research, we select 27 EEG channels for emotion recognition experiments based on the attention weight distribution and electrode symmetry in Fig. 8, namely FPZ, FP1, FP2, AF3, AF4, F7, F8, FT7, FT8, T7, T8, TP7, TP8, C3, C4, PO3, PO4, P7, P5, P3, P4, P6, P8, PO5, PO6, PO7 and PO8 channels. Fig. 9 describes the experimental effect of 27 EEG channels and 62 EEG channels for subject-independent emotion experiment. As can be seen from Fig. 9, the accuracy of 27 electrodes are 69.07%, 91.48%, 98.33%, 95.19%, 99.81%, 99.63%, 100%,

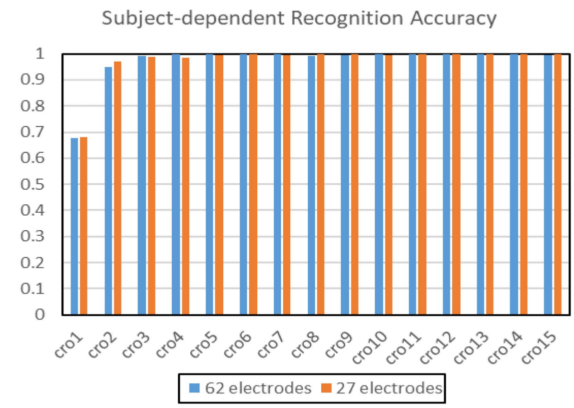

Fig. 10. Recognition results of subject-dependent experiment with different EEG channels.

99.81%, 99.81%, 100%, 99.63%, 100%, 98.7%, 100%, 99.81%; The accuracy of 62 electrodes are 62.96%, 89.81%, 97.78%, 98.89%, 99.07%, 99.44%, 99.26%, 100%, 100%, 99.44%, 100%, 99.26%, 99.81%, 100%, 99.81%. The average accuracy of subject-independent experiments with 27 EEG channels and 62 EEG channels is 96.75% and 96.37%, respectively. The 27 EEG channels can not only improve the accuracy of some subjects but also improve the average accuracy of all subjects. This is mainly because the highly correlated EEG channels not only cut down feature redundancy but also cut down the impact of noise, so as to be more conducive to emotion classification.

Similarly, Fig. 10 shows the experimental results of 27 EEG channels and 62 EEG channels for subject-dependent emotion experiment. The accuracy of 27 electrodes are 68.15%, 97.04%, 98.7%, 98.33%, 99.63%, 99.81%, 99.44%, 100%, 100%, 99.63%, 99.81%, 100%, 100%, 100%, 100%; The accuracy of 62 electrodes are 67.78%, 95%, 99.26%, 100%, 100%, 99.81%, 100%, 99.07%, 99.63%, 100%, 100%, 100%, 100%, 100%, 99.63%. The average accuracy of the 15-fold cross-validation experiments on 27 EEG channels and 62 EEG channels are 97.37% and 97.35%, respectively. Fig. 10 illustrates although the average accuracy of the 27 EEG channels is not much different from that of the 62 EEG channels, the reduction of the number of EEG channels can reduce the network computation and provide a theoretical basis for the innovation of portable EEG equipment.

V. Conclusion

In this study, we illustrate an EEG research based on three-dimension convolution attention neural network, which effectively optimizes the performance of emotion recognition. The 3DCANN method includes the 3DCNN feature extraction part and channels attention weight learning part. Firstly, the 3D convolution in the 3DCNN module not only calculates features in the spatial dimension but also extracts dynamic information in the time domain. It can automatically learn the complex features contained in the EEG image and show good performance in biosignal recognition tasks. The 3D max-pooling greatly cuts down the dimension of the feature map and effectively prevents the occurrence of over-fitting. Secondly, to cut down feature redundancy and study the relationship between EEG channels and emotional states, we use the AM block to learn the channel attention weight matrix. Specifically, the attention weight distribution of 62 electrodes is obtained through the softmax activation function and it is mapped to the brain scalp, which explains the connection between various parts of the brain and emotions. Finally, according to the evaluation results of the algorithm, 3DCANN optimizes the EEGs' characteristic information and exploits hidden spatio-temporal characteristics to achieve the best performance. The proposed method has certain application significance in human-computer affective interaction and psychology research.

In addition, we select 27 EEG channels that are highly related to EEG emotion recognition through the attention weight distribution diagram. The experimental results show that the use of 27 EEG channels can not only reduce the influence of noise on the recognition effect but also improve the discomfort of wearable EEG devices. Finally, we will conduct in-depth exploration in the EEG emotion research, optimize the stability [45]–[47] and accuracy of our algorithm and provide valuable experience for the realization of a new generation of more secure and intelligent human-computer interaction technology.

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